PV101 Final Assessment Answer Key

Instructor copy | 25 questions | 25 points

Day_08_PV101_Final_Review_and_Assessment_2026-07-16

Q	Chapter	Type	Answer / acceptable evidence
1	1	MC	B. Light
2	1	MC	C. An off-grid cabin used at night
3	2	MC	B. Isolate energy sources and verify the state
4	2	MC	C. Stop and verify with the correct meter setup
5	3	MC	A. Voltage is electrical pressure and current is flow
6	3	CALC	36 W; W = V x A = 12 x 3.
7	4	MC	A. Daily equivalent full-sun energy
8	4	MC	B. Output is reduced
9	5	MC	B. A possible DC voltage source
10	5	DIAGRAM	Voltage adds; current remains limited by the matched module/string current.
11	6	MC	B. Regulate charging to protect the battery
12	6	MC	A. Convert DC to AC
13	7	CALC	120 Wh/day; 3 x 10 W x 4 h.

14	7	MC	A. A load assessment
15	8	MC	B. To protect as much conductor as practical
16	8	MC	B. Voltage drop and heating
17	9	MC	B. It can be easier to access and orient
18	9	DIAGRAM	Ground area; it has better documented solar access. Accept reason tied to shading/access, with acknowledgment that full site review is still required.
19	10	MC	B. Choose one test that can separate likely causes
20	10	SHORT	The cause and conductor/system condition must be checked, then the known-good state must be verified with repeat measurements or functional testing.
21	11	MC	C. Enforces the applicable local requirements and approvals
22	11	CALC	$2,000; $2,500 - $500 = $2,000.
23	12	SHORT	Reviewing misses identifies the specific concept, calculation, unit, or safety reasoning that needs correction and turns the practice set into a learning tool.
24	12	CALC	750 W; 0.75 kW x 1,000 W/kW.

25	Cumulative	SCENARIO	Accept: stop and isolate energy; inspect/trace the actual path and protection; verify meter setup; with the circuit de-energized test fuse continuity or measure voltage at logical points after safe re-energization; record expected vs actual readings, conclusion, repair, and restoration check. Must not bypass protection or trust the label alone.

Scoring

Award one point per question. For calculations, require the correct result with an appropriate unit; accept equivalent work. For written items, accept equivalent safe reasoning that includes the listed evidence. Record the raw score out of 25.